WORKED SOLUTION

STEP I 2014, Question 5

Pure • Cubic Inequalities

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q5

The key idea

Part (i) is a single-variable cubic whose minimum on $x \geq 0$ turns out to be exactly zero, touching the axis at $x = a$. That one inequality, $(x+2a)^3 \geq 27a^2x$, is an AM–GM inequality in disguise: the left side involves the sum $x + a + a$ and the right side the product $x \cdot a \cdot a$. Parts (ii) and (iii) are about choosing what to call x and a, and tracking exactly when equality holds.

Solution

Part (i)

Differentiate:

$$f'(x) = 3(x+2a)^2 - 27a^2 = 3((x+2a)^2 - 9a^2) = 3(x-a)(x+5a),$$

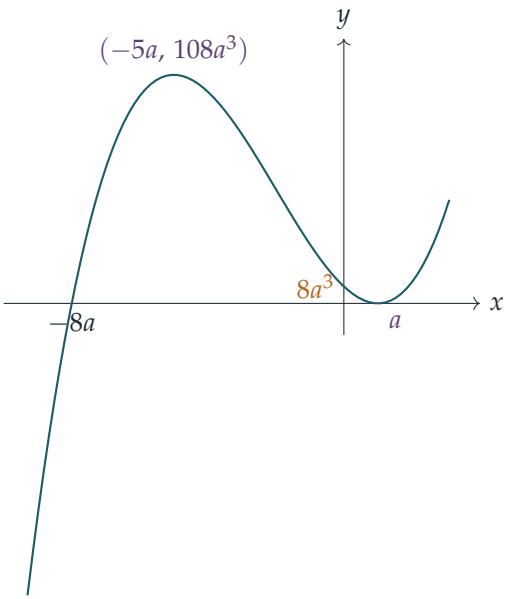
so the stationary points are at $x = a$ and $x = -5a$. There

$$f(a) = (3a)^3 - 27a^3 = 0, \qquad f(-5a) = (-3a)^3 + 135a^3 = 108a^3,$$

and $f(0) = 8a^3$. Since f is a cubic with positive leading coefficient, $x = -5a$ is a local maximum and $x = a$ a local minimum. Because the minimum value is 0, the graph touches the x-axis at $x = a$, so $(x-a)^2$ is a factor; comparing constant terms then gives

$$f(x) = (x-a)^2(x+8a),$$

which you can confirm by expanding both sides to $x^3 + 6ax^2 - 15a^2x + 8a^3$.



For $x \geq 0$ the graph starts at height $8a^3 \geq 0$, decreases to 0 at $x = a$, and then increases for ever. So the least value of f on $x \geq 0$ is 0, and

$$f(x) \geq 0 \quad \text{for } x \geq 0,$$

with equality only at $x = a$. (The factorised form says the same thing directly: for $x \geq 0$, $(x-a)^2 \geq 0$ and $x + 8a > 0$ unless $x = a = 0$.) If $a = 0$ the picture degenerates to $f(x) = x^3$, which is still non-negative for $x \geq 0$, with equality only at $x = 0 = a$. In every case

$$(x+2a)^3 \geq 27a^2x \quad (x, a \geq 0), \qquad \text{with equality if and only if } x = a.$$

Part (ii)

We want xy^2, so take $a = y \geq 0$ in part (i). For $x \geq 0$, $y \geq 0$,

$$27xy^2 \leq (x+2y)^3 \leq 3^3 = 27,$$

the second step because $0 \leq x + 2y \leq 3$ and cubing is increasing. Hence $xy^2 \leq 1$ throughout the region.

The bound is attained: equality needs $x = y$ (from part (i)) and $x + 2y = 3$, which give $x = y = 1$, and indeed $1 \cdot 1^2 = 1$. So the greatest value of xy^2 is 1, achieved only at $x = 1$, $y = 1$.

Part (iii)

Ask yourself Part (i) involves only two numbers, x and a. How can it say anything about three?

Group q and r together: put $x = p$ and $2a = q + r$, so $a = \frac{1}{2}(q + r) \geq 0$. Part (i) gives

$$(p + q + r)^3 \geq 27p\left(\frac{q + r}{2}\right)^2.$$

Now $\left(\frac{q+r}{2}\right)^2 - qr = \frac{(q-r)^2}{4} \geq 0$, and $p \geq 0$, so

$$(p + q + r)^3 \geq 27p\left(\frac{q + r}{2}\right)^2 \geq 27pqr,$$

which is the required inequality.

Equality. If $(p + q + r)^3 = 27pqr$ then both inequalities in the chain are equalities.

- The first is equality exactly when $x = a$, that is $p = \frac{1}{2}(q + r)$.
- The second is equality exactly when $\frac{27}{4}p(q - r)^2 = 0$, that is $p = 0$ or $q = r$.

If $p = 0$, the first condition gives $q + r = 0$, so $q = r = 0$ as well. If $q = r$, the first condition gives $p = q$. Either way $p = q = r$. Conversely, if $p = q = r$ then both sides equal $27p^3$. So equality holds if and only if

$$p = q = r.$$

Checks

- Expanding $(x - a)^2(x + 8a)$ recovers $(x + 2a)^3 - 27a^2x$; with $a = 1$, $f(0) = 8$, $f(1) = 0$, $f(3) = 4 \cdot 11 = 44$, all non-negative.
- In part (ii), try other points on the line $x + 2y = 3$: $(x, y) = (2, \frac{1}{2})$ gives $\frac{1}{2}$, and $(0.5, 1.25)$ gives about 0.78, both below 1. A computer search over a fine grid of the triangle also gives a maximum of 1 at $(1, 1)$.
- In part (iii), $p = 1, q = 2, r = 3$ gives $216 \geq 162$; a computer search over small whole numbers found equality only when $p = q = r$.

Common slips

- Showing $f(a) = 0$ but not explaining why this is the *least* value on $x \geq 0$: you need the shape of the graph (or the factorisation), including the endpoint value $f(0) = 8a^3 \geq 0$.
- In part (ii), forgetting to check that the bound 1 is actually reached; an upper bound alone is not a greatest value.

- In the equality case of part (iii), concluding only $q = r$ and missing the alternative $p = 0$, which has to be ruled out (or shown to force $p = q = r = 0$).

Beyond STEP

Part (iii) is the three-term AM–GM inequality, $\frac{1}{3}(p+q+r) \geq \sqrt[3]{pqr}$, and the method here, merging two variables into their average and using a two-variable inequality, is the germ of the classical proofs of AM–GM for n terms. Part (ii) is a small constrained optimisation problem, the sort that university courses attack with Lagrange multipliers, though inequalities like AM–GM often give the answer far more cleanly. For much more in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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